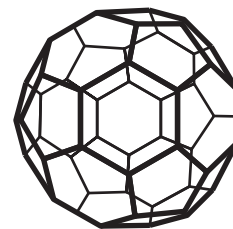


Introduction to Inorganic Chemistry



1.1 What Is Inorganic Chemistry?

If organic chemistry is defined as the chemistry of hydrocarbon compounds and their derivatives, inorganic chemistry can be described broadly as the chemistry of “everything else.” This includes all the remaining elements in the periodic table, as well as carbon, which plays a major and growing role in inorganic chemistry. The large field of organo-metallic chemistry bridges both areas by considering compounds containing metal–carbon bonds; it also includes catalysis of many organic reactions. Bioinorganic chemistry bridges biochemistry and inorganic chemistry and has an important focus on medical applications. Environmental chemistry includes the study of both inorganic and organic compounds. In short, the inorganic realm is vast, providing essentially limitless areas for investigation and potential practical applications.

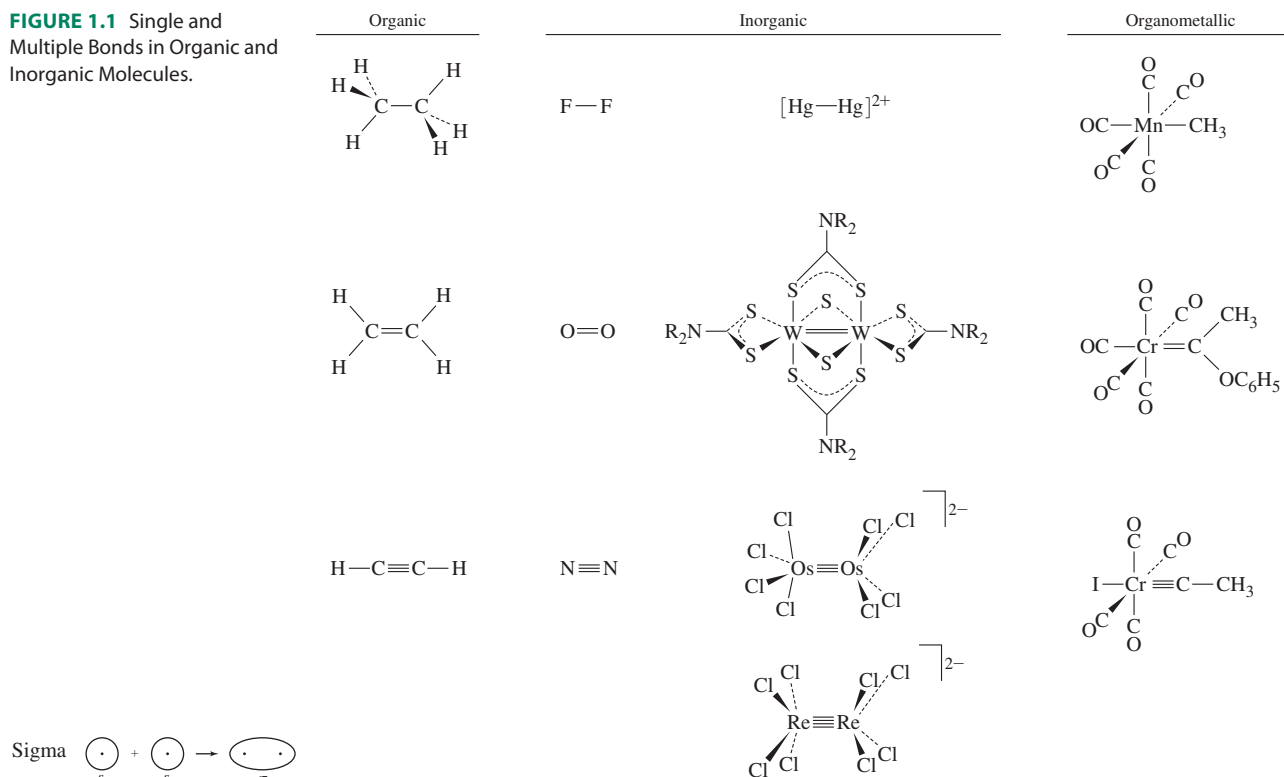
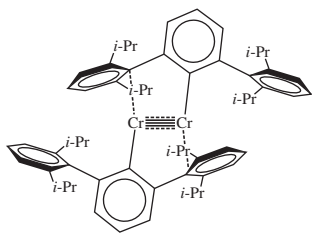
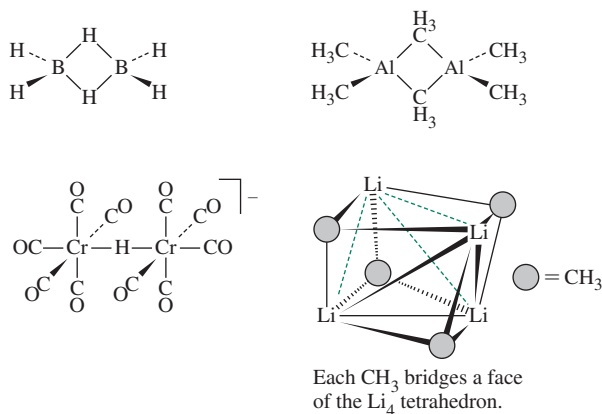
1.2 Contrasts with Organic Chemistry

Some comparisons between organic and inorganic compounds are in order. In both areas, single, double, and triple covalent bonds are found (**Figure 1.1**); for inorganic compounds, these include direct metal–metal bonds and metal–carbon bonds. Although the maximum number of bonds between two carbon atoms is three, there are many compounds that contain quadruple bonds between metal atoms. In addition to the sigma and pi bonds common in organic chemistry, quadruply bonded metal atoms contain a delta (δ) bond (**Figure 1.2**); a combination of one sigma bond, two pi bonds, and one delta bond makes up the quadruple bond. The delta bond is possible in these cases because the metal atoms have *d* orbitals to use in bonding, whereas carbon has only *s* and *p* orbitals energetically accessible for bonding.

Compounds with “fivefold” bonds between transition metals have been reported (**Figure 1.3**), accompanied by debate as to whether these bonds merit the designation “quintuple.”

In organic compounds, hydrogen is nearly always bonded to a single carbon. In inorganic compounds, hydrogen is frequently encountered as a bridging atom between two or more other atoms. Bridging hydrogen atoms can also occur in metal cluster compounds, in which hydrogen atoms form bridges across edges or faces of polyhedra of metal atoms. Alkyl groups may also act as bridges in inorganic compounds, a function rarely encountered in organic chemistry except in reaction intermediates. Examples of terminal and bridging hydrogen atoms and alkyl groups in inorganic compounds are in **Figure 1.4**.

Some of the most striking differences between the chemistry of carbon and that of many other elements are in coordination number and geometry. Although carbon is usually limited to a maximum coordination number of four (a maximum of four atoms bonded

FIGURE 1.1 Single and Multiple Bonds in Organic and Inorganic Molecules.**FIGURE 1.2** Examples of Bonding Interactions.**FIGURE 1.3** Example of Fivefold Bonding.**FIGURE 1.4** Examples of Inorganic Compounds Containing Terminal and Bridging Hydrogens and Alkyl Groups.

to carbon, as in CH_4), numerous inorganic compounds have central atoms with coordination numbers of five, six, seven, and higher; the most common coordination geometry for transition metals is an octahedral arrangement around a central atom, as shown for $[TiF_6]^{3-}$ (**Figure 1.5**). Furthermore, inorganic compounds present coordination geometries different from those found for carbon. For example, although 4-coordinate carbon is nearly always tetrahedral, both tetrahedral and square-planar shapes occur for 4-coordinate compounds of both metals and nonmetals. When metals are in the center, with anions or neutral molecules (*ligands*) bonded to them (frequently through N, O, or S), these are called *coordination complexes*; when carbon is the element directly bonded to metal atoms or ions, they are also classified as *organometallic complexes*.

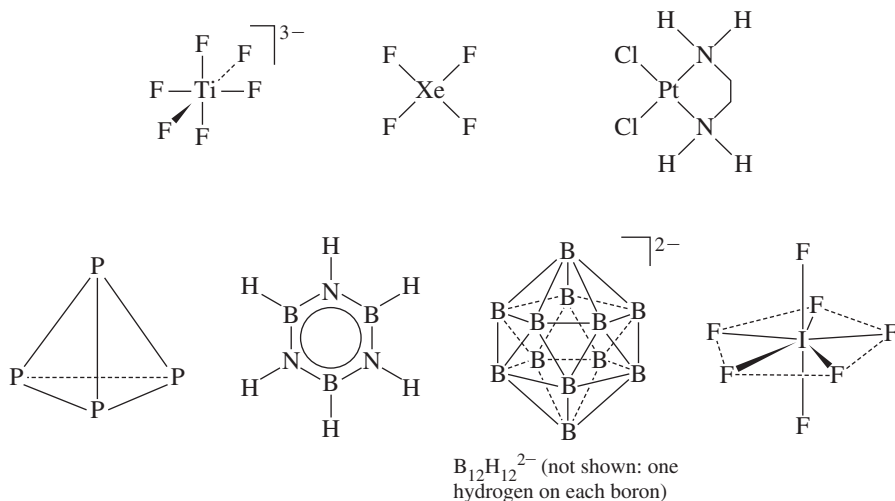


FIGURE 1.5 Examples of Geometries of Inorganic Compounds.

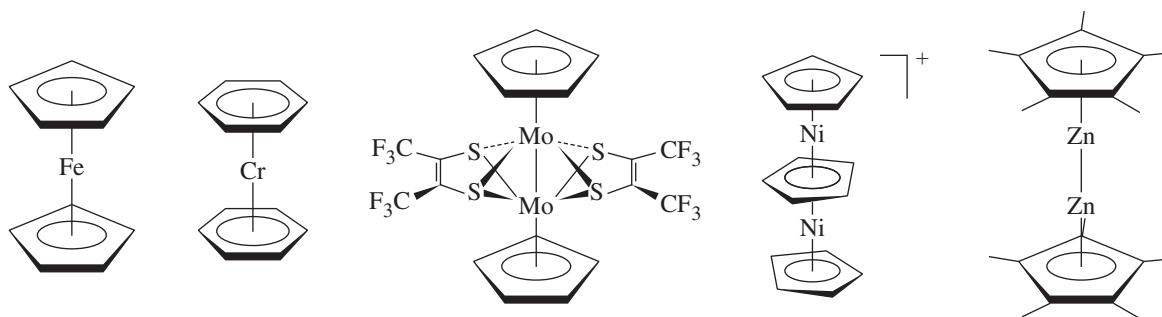


FIGURE 1.6 Inorganic Compounds Containing Pi-Bonded Aromatic Rings.

The tetrahedral geometry usually found in 4-coordinate compounds of carbon also occurs in a different form in some inorganic molecules. Methane contains four hydrogens in a regular tetrahedron around carbon. Elemental phosphorus is tetratomic (P_4) and tetrahedral, but with no central atom. Other elements can also form molecules in which outer atoms surround a central cavity; an example is boron, which forms numerous structures containing icosahedral B_{12} units. Examples of some of the geometries found for inorganic compounds are in Figure 1.5.

Aromatic rings are common in organic chemistry, and aryl groups can also form sigma bonds to metals. However, aromatic rings can also bond to metals in a dramatically different fashion using their pi orbitals, as shown in Figure 1.6 and in this book's cover illustration. The result is a metal atom bonded above the center of the ring, almost as if suspended in space. In many cases, metal atoms are sandwiched between two aromatic rings. Multiple-decker sandwiches of metals and aromatic rings are also known.

Carbon plays an unusual role in a number of metal *cluster compounds* in which a carbon atom is at the center of a polyhedron of metal atoms. Examples of carbon-centered clusters with five, six, or more surrounding metals are known (Figure 1.7). The striking role that carbon plays in these clusters has provided a challenge to theoretical inorganic chemists.

In addition, since the mid-1980s the chemistry of elemental carbon has flourished. This phenomenon began with the discovery of fullerenes, most notably the cluster C_{60} , dubbed “buckminsterfullerene” after the developer of the geodesic dome. Many other fullerenes (buckyballs) are now known and serve as cores of a variety of derivatives. In

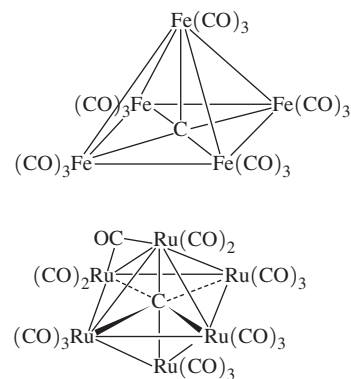
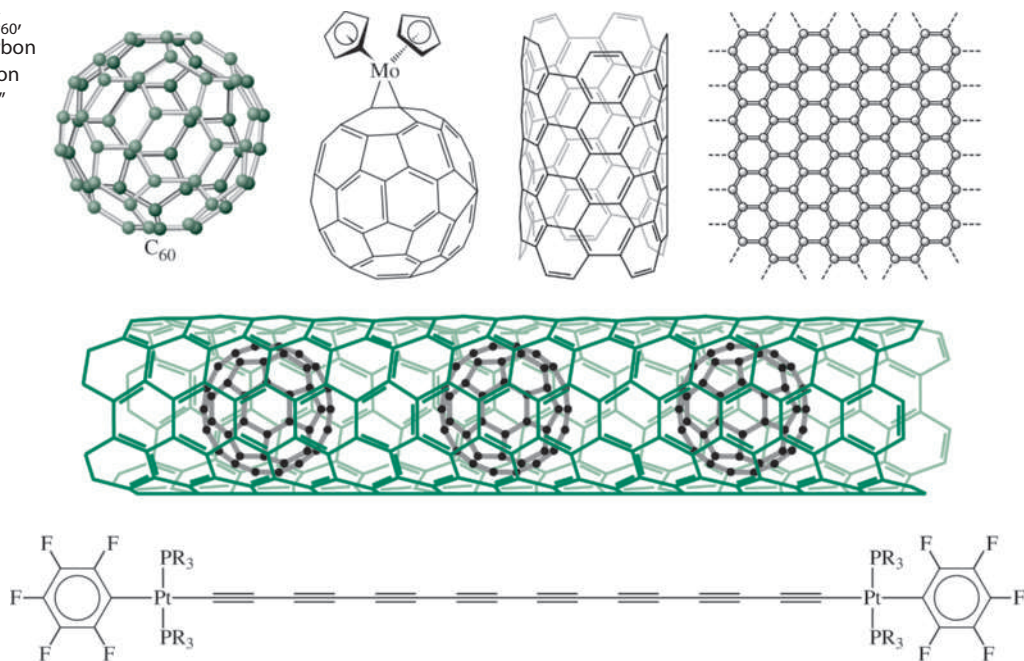


FIGURE 1.7 Carbon-Centered Metal Clusters.

FIGURE 1.8 The Fullerene C_{60} , a Fullerene Compound, a Carbon Nanotube, Graphene, a Carbon Peapod, and a Polyyne “Wire” Connecting Platinum Atoms.



addition, numerous other forms of carbon (for example, carbon nanotubes, nanoribbons, graphene, and carbon wires) have attracted much interest and show potential for applications in fields as diverse as nanoelectronics, body armor, and drug delivery. **Figure 1.8** provides examples of these newer forms of carbon.

The era of sharp dividing lines between subfields in chemistry has long been obsolete. Many of the subjects in this book, such as acid–base chemistry and organometallic reactions, are of vital interest to organic chemists. Other topics such as oxidation–reduction reactions, spectra, and solubility relations interest analytical chemists. Subjects related to structure determination, spectra, conductivity, and theories of bonding appeal to physical chemists. Finally, the use of organometallic catalysts provides a connection to petroleum and polymer chemistry, and coordination compounds such as hemoglobin and metal-containing enzymes provide a similar tie to biochemistry. Many inorganic chemists work with professionals in other fields to apply chemical discoveries to addressing modern challenges in medicine, energy, the environment, materials science, and other fields. In brief, modern inorganic chemistry is not a fragmented field of study, but has numerous interconnections with other fields of science, medicine, technology, and other disciplines.

The remainder of this chapter is devoted to a short history of the origins of inorganic chemistry and perspective on more recent developments, intended to provide a sense of connection to the past and to place some aspects of inorganic chemistry within the context of larger historical events. In later chapters, brief historical context is provided with the same intention.

1.3 The History of Inorganic Chemistry

Even before alchemy became a subject of study, many chemical reactions were used and their products applied to daily life. The first metals used were probably gold and copper, which can be found in the metallic state in nature. Copper can also be readily formed by the reduction of malachite—basic copper carbonate, $Cu_2(CO_3)(OH)_2$ —in charcoal fires. Silver, tin, antimony, and lead were also known as early as 3000 BCE. Iron appeared in

classical Greece and in other areas around the Mediterranean Sea by 1500 BCE. At about the same time, colored glasses and ceramic glazes were introduced, largely composed of silicon dioxide (SiO_2 , the major component of sand) and other metallic oxides, which had been melted and allowed to cool to amorphous solids.

Alchemists were active in China, Egypt, and other centers of civilization early in the first centuries CE. Although much effort went into attempts to “transmute” base metals into gold, alchemists also described many other chemical reactions and operations. Distillation, sublimation, crystallization, and other techniques were developed and used in their studies. Because of the political and social changes of the time, alchemy shifted into the Arab world and later—about 1000 to 1500 CE—reappeared in Europe. Gunpowder was used in Chinese fireworks as early as 1150, and alchemy was also widespread in China and India at that time. Alchemists appeared in art, literature, and science until at least 1600, by which time chemistry was beginning to take shape as a science. Roger Bacon (1214–1294), recognized as one of the first great experimental scientists, also wrote extensively about alchemy.

By the seventeenth century, the common strong acids—nitric, sulfuric, and hydrochloric—were known, and systematic descriptions of common salts and their reactions were being accumulated. As experimental techniques improved, the quantitative study of chemical reactions and the properties of gases became more common, atomic and molecular weights were determined more accurately, and the groundwork was laid for what later became the periodic table of the elements. By 1869, the concepts of atoms and molecules were well established, and it was possible for Mendeleev and Meyer to propose different forms of the periodic table. **Figure 1.9** illustrates Mendeleev’s original periodic table.*

The chemical industry, which had been in existence since very early times in the form of factories for purifying salts and for smelting and refining metals, expanded as methods for preparing relatively pure materials became common. In 1896, Becquerel discovered radioactivity, and another area of study was opened. Studies of subatomic particles, spectra, and electricity led to the atomic theory of Bohr in 1913, which was soon modified by the quantum mechanics of Schrödinger and Heisenberg in 1926 and 1927.

Inorganic chemistry as a field of study was extremely important during the early years of the exploration and development of mineral resources. Qualitative analysis methods were

			Ti = 50	Zr = 90	? = 180
			V = 51	Nb = 94	Ta = 182
			Cr = 52	Mo = 96	W = 186
			Mn = 53	Rh = 104.4	Pt = 197.4
			Fe = 56	Ru = 104.2	Ir = 198
			Ni = Co = 59	Pd = 106.6	Os = 199
			Cu = 63.4	Ag = 108	Hg = 200
			Zn = 65.2	Cd = 112	
			? = 68	Ur = 116	Au = 197?
			? = 70	Sn = 118	
			As = 75	Sb = 122	Bi = 210?
			Se = 79.4	Te = 128?	
			Br = 80	J = 127	
			Rb = 85.4	Cs = 133	Tl = 204
			Sr = 87.6	Ba = 137	Pb = 207
			? = 45	Ce = 92	
			?Er = 56	La = 94	
			?Yt = 60	Di = 95	
			?In = 75.6	Th = 118 ?	
H = 1	Be = 9.4	Mg = 24			
	B = 11	Al = 27.4			
	C = 12	Si = 28			
	N = 14	P = 31			
	O = 16	S = 32			
	F = 19	Cl = 35.5			
Li = 7	Na = 23	K = 39			
		Ca = 40			

FIGURE 1.9 Mendeleev’s 1869 Periodic Table. Two years later, Mendeleev revised his table into a form similar to a modern short-form periodic table, with eight groups across.

*The original table was published in *Zeitschrift für Chemie*, **1869**, 12, 405. It can be found in English translation, together with a page from the German article, at web.lemoyne.edu/~giunta/mendeleev.html. See M. Laing, *J. Chem. Educ.*, **2008**, 85, 63 for illustrations of Mendeleev’s various versions of the periodic table, including his handwritten draft of the 1869 table.

developed to help identify minerals and, combined with quantitative methods, to assess their purity and value. As the Industrial Revolution progressed, so did the chemical industry. By the early twentieth century, plants for the high volume production of ammonia, nitric acid, sulfuric acid, sodium hydroxide, and many other inorganic chemicals were common.

Early in the twentieth century, Werner and Jørgensen made considerable progress on understanding the coordination chemistry of transition metals and also discovered a number of organometallic compounds. Nevertheless, the popularity of inorganic chemistry as a field of study gradually declined during most of the first half of the century. The need for inorganic chemists to work on military projects during World War II rejuvenated interest in the field. As work was done on many projects (not least of which was the Manhattan Project, in which scientists developed the fission bomb), new areas of research appeared, and new theories were proposed that prompted further experimental work. A great expansion of inorganic chemistry began in the 1940s, sparked by the enthusiasm and ideas generated during World War II.

In the 1950s, an earlier method used to describe the spectra of metal ions surrounded by negatively charged ions in crystals (**crystal field theory**)¹ was extended by the use of molecular orbital theory² to develop **ligand field theory** for use in coordination compounds, in which metal ions are surrounded by ions or molecules that donate electron pairs. This theory gave a more complete picture of the bonding in these compounds. The field developed rapidly as a result of this theoretical framework, availability of new instruments, and the generally reawakened interest in inorganic chemistry.

In 1955, Ziegler³ and Natta⁴ discovered organometallic compounds that could catalyze the polymerization of ethylene at lower temperatures and pressures than the common industrial method at that time. In addition, the polyethylene formed was more likely to be made up of linear, rather than branched, molecules and, as a consequence, was stronger and more durable. Other catalysts were soon developed, and their study contributed to the rapid expansion of organometallic chemistry, still a rapidly growing area.

The study of biological materials containing metal atoms has also progressed rapidly. The development of new experimental methods allowed more thorough study of these compounds, and the related theoretical work provided connections to other areas of study. Attempts to make *model* compounds that have chemical and biological activity similar to the natural compounds have also led to many new synthetic techniques. Two of the many biological molecules that contain metals are in **Figure 1.10**. Although these molecules have very different roles, they share similar ring systems.

One current area that bridges organometallic chemistry and bioinorganic chemistry is the conversion of nitrogen to ammonia:



This reaction is one of the most important industrial processes, with over 100 million tons of ammonia produced annually worldwide, primarily for fertilizer. However, in spite of metal oxide catalysts introduced in the Haber–Bosch process in 1913, and improved since then, it is also a reaction that requires temperatures between 350 and 550 °C and from 150–350 atm pressure and that still results in a yield of only 15 percent ammonia. Bacteria, however, manage to fix nitrogen (convert it to ammonia and then to nitrite and nitrate) at 0.8 atm at room temperature in nodules on the roots of legumes. The nitrogenase enzyme that catalyzes this reaction is a complex iron–molybdenum–sulfur protein. The structure of its active sites has been determined by X-ray crystallography.⁵ A vigorous area of modern inorganic research is to design reactions that could be carried out on an industrial scale that model the reaction of nitrogenase to generate ammonia under mild conditions. It is estimated that as much as 1 percent of the world's total energy consumption is currently used for the Haber–Bosch process.

Inorganic chemistry also has medical applications. Notable among these is the development of platinum-containing antitumor agents, the first of which was the *cis* isomer of $\text{Pt}(\text{NH}_3)_2\text{Cl}_2$,

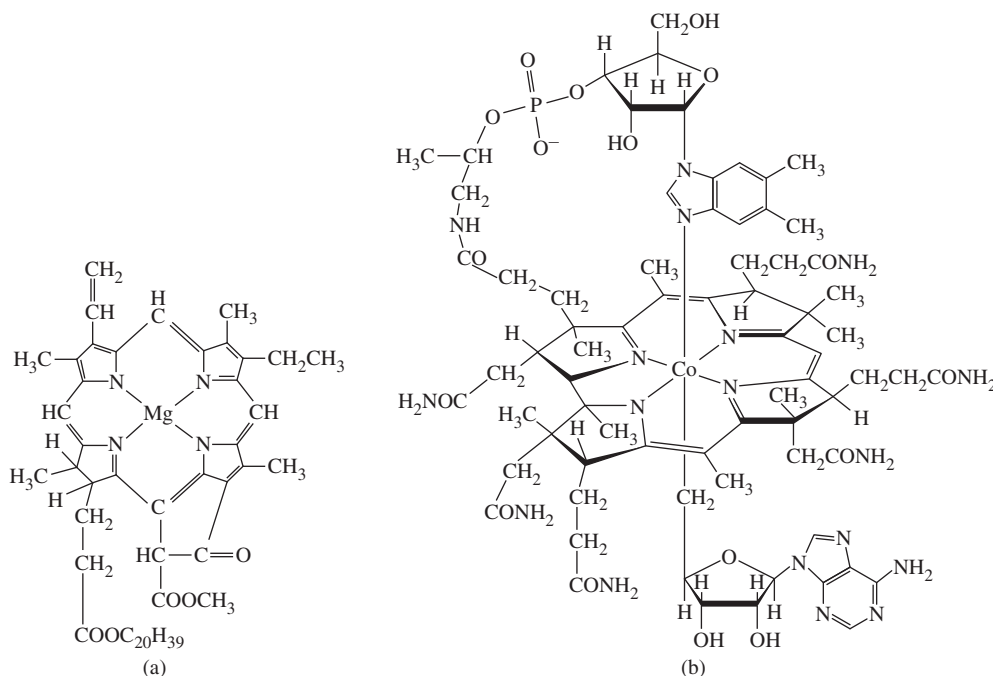


FIGURE 1.10 Biological Molecules Containing Metal Ions. (a) Chlorophyll a, the active agent in photosynthesis. (b) Vitamin B₁₂ coenzyme, a naturally occurring organometallic compound.

cisplatin. First approved for clinical use approximately 30 years ago, cisplatin has served as the prototype for a variety of anticancer agents; for example, satraplatin, the first orally available platinum anticancer drug to reach clinical trials.* These two compounds are in Figure 1.11.

1.4 Perspective

The premier issue of the journal *Inorganic Chemistry*** was published in February 1962. Much of the focus of that issue was on classic coordination chemistry, with more than half its research papers on synthesis of coordination complexes and their structures and properties. A few papers were on compounds of nonmetals and on organometallic chemistry, then a relatively new field; several were on thermodynamics or spectroscopy. All of these topics have developed considerably in the subsequent half-century, but much of the evolution of inorganic chemistry has been into realms unforeseen in 1962.

The 1962 publication of the first edition of F. A. Cotton and G. Wilkinson's landmark text *Advanced Inorganic Chemistry*⁶ provides a convenient reference point for the status of inorganic chemistry at that time. For example, this text cited only the two long-known forms of carbon, diamond and graphite, although it did mention "amorphous forms" attributed to microcrystalline graphite. It would not be until more than two decades later that carbon chemistry would explode with the seminal discovery of C₆₀ in 1985 by Kroto, Curl, Smalley, and colleagues,⁷ followed by other fullerenes, nanotubes, graphene, and other forms of carbon (Figure 1.8) with the potential to have major impacts on electronics, materials science, medicine, and other realms of science and technology.

As another example, at the beginning of 1962 the elements helium through radon were commonly dubbed "inert" gases, believed to "form no chemically bound compounds" because of the stability of their electron configurations. Later that same year, Bartlett

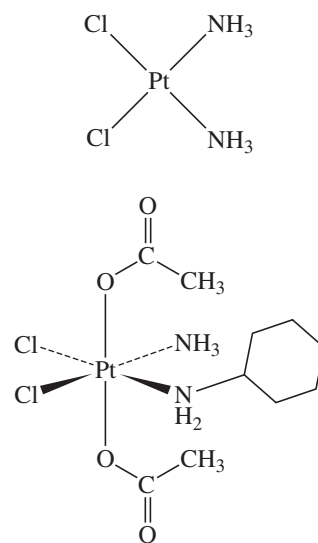


FIGURE 1.11 Cisplatin and Satraplatin.

*For reviews of modes of interaction of cisplatin and related drugs, see P. C. A. Bruijninx, P. J. Sadler, *Curr. Opin. Chem. Bio.*, **2008**, 12, 197 and F. Arnesano, G. Natile, *Coord. Chem. Rev.*, **2009**, 253, 2070.

**The authors of this issue of *Inorganic Chemistry* were a distinguished group, including five recipients of the Priestley Medal, the highest honor conferred by the American Chemical Society, and 1983 Nobel Laureate Henry Taube.

reported the first chemical reactions of xenon with PtF_6 , launching the synthetic chemistry of the now-renamed “noble” gas elements, especially xenon and krypton;⁸ numerous compounds of these elements have been prepared in succeeding decades.

Numerous square planar platinum complexes were known by 1962; the chemistry of platinum compounds had been underway for more than a century. However, it was not known until Rosenberg’s work in the latter part of the 1960s that one of these, *cis*- $\text{Pt}(\text{NH}_3)_2\text{Cl}_2$ (cisplatin, Figure 1.11), had anticancer activity.⁹ Antitumor agents containing platinum and other transition metals have subsequently become major tools in treatment regimens for many types of cancer.¹⁰

That first issue of *Inorganic Chemistry* contained only 188 pages, and the journal was published quarterly, exclusively in hardcopy. Researchers from only four countries were represented, more than 90 percent from the United States, the others from Europe. *Inorganic Chemistry* now averages approximately 550 pages per issue, is published 24 times annually, and publishes (electronically) research conducted broadly around the globe. The growth and diversity of research published in *Inorganic Chemistry* has been paralleled in a wide variety of other journals that publish articles on inorganic and related fields.

In the preface to the first edition of *Advanced Inorganic Chemistry*, Cotton and Wilkinson stated, “in recent years, inorganic chemistry has experienced an impressive renaissance.” This renaissance shows no sign of diminishing.

With this brief survey of the marvelously complex field of inorganic chemistry, we now turn to the details in the remainder of this book. The topics included provide a broad introduction to the field. However, even a cursory examination of a chemical library or one of the many inorganic journals shows some important aspects of inorganic chemistry that must be omitted in a textbook of moderate length. The references cited in this text suggest resources for further study, including historical sources, texts, and reference works that provide useful additional material.

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General References

For those who are interested in the historical development of inorganic chemistry focused on metal coordination compounds during the period 1798–1935, copies of key research papers, including translations, are provided in the three-volume set *Classics in Coordination Chemistry*, G. B. Kauffman, ed., Dover Publications, N.Y. 1968, 1976, 1978. Among the many general reference works available, three of the most useful and complete are N. N. Greenwood and A. Earnshaw’s *Chemistry of*

the Elements, 2nd ed., Butterworth-Heinemann, Oxford, 1997; F. A. Cotton, G. Wilkinson, C. A. Murillo, and M. Bochman’s *Advanced Inorganic Chemistry*, 6th ed., John Wiley & Sons, New York, 1999; and A. F. Wells’s *Structural Inorganic Chemistry*, 5th ed., Oxford University Press, New York, 1984. An interesting study of inorganic reactions from a different perspective can be found in G. Wulfsberg’s *Principles of Descriptive Inorganic Chemistry*, Brooks/Cole, Belmont, CA, 1987.